

Chapter 46

Animal Reproduction

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

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Overview: Pairing Up for Sexual Reproduction

- Each earthworm produces sperm and eggs; in a few weeks, new worms will hatch from fertilized eggs.
- Animal reproduction takes many forms.
- Aspects of animal form and function can be viewed broadly as adaptations contributing to reproductive success.

How can each of these earthworms be both male and female?



Both asexual and sexual reproduction occur in the animal kingdom

- **Sexual reproduction** is the creation of an offspring by fusion of a male gamete (*sperm*) and female gamete (*egg*) to form a *zygote*.
- **Asexual reproduction** is creation of offspring without the fusion of egg and sperm . One parent *clones* offspring.
- Many invertebrates reproduce asexually by **fission** = *separation of a parent into two or more individuals of about the same size.*

Asexual reproduction of a sea anemone



Asexual Reproduction

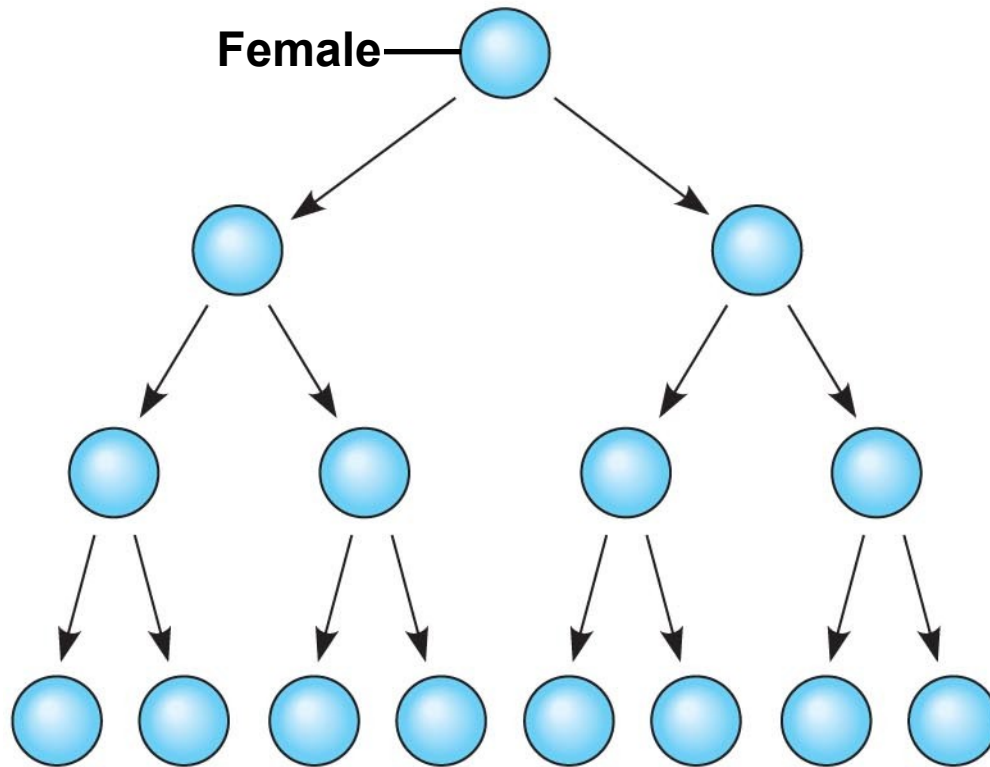
- **Budding** = new individuals arise from outgrowths of existing ones.
- **Fragmentation** = breaking of the body into pieces, some or all of which develop into adults.
- *Fragmentation must be accompanied by regeneration* = regrowth of lost body parts.
- **Parthenogenesis** is the *development of a new individual from an unfertilized egg*.

Sexual Reproduction: An Evolutionary Enigma

- Sexual females have half as many daughters as asexual females; this is the “twofold cost” of sexual reproduction.
- Despite this, almost all eukaryotic species reproduce sexually.

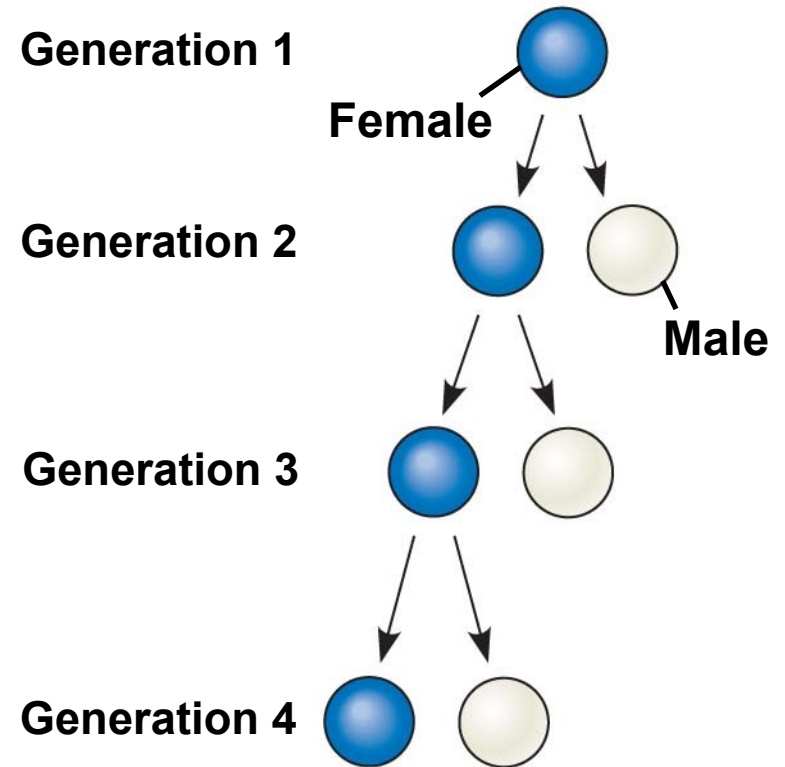
The “reproductive handicap” of sex: Sexual females have half as many daughters as asexual females.

Asexual reproduction



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Sexual reproduction



Sexual reproduction - Variety

- Sexual reproduction results in genetic recombination, which provides potential advantages:
 - An increase in variation in offspring, providing an increase in the reproductive success of parents in changing environments
 - An increase in the rate of adaptation
 - A shuffling of genes and the elimination of harmful genes from a population.

Reproductive Cycles and Patterns

- **Ovulation** is the release of mature eggs at the midpoint of a female cycle.
- Most animals exhibit **reproductive cycles** related to changing **seasons**.
- Reproductive cycles are controlled by **hormones** and **environmental cues**.
- Animals may reproduce asexually or sexually, or they may alternate these methods.

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- Sexual reproduction is a special problem for organisms that seldom encounter a mate.
 - One solution is **hermaphroditism** = *each individual has BOTH male and female reproductive systems.*
 - Some hermaphrodites can self-fertilize.

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- Individuals of some species undergo **sex reversals**.
 - Some species exhibit male to female reversal (for example, certain oysters), while others exhibit female to male reversal (for example, a coral reef fish).

Fertilization depends on mechanisms that bring together sperm and eggs of the same species

- The **mechanisms** of **fertilization**, the union of egg and sperm, play an important part in sexual reproduction.
- In **external fertilization**, eggs shed by the female are fertilized by sperm in the external environment.

External fertilization



Eggs

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- In **internal fertilization**, sperm are deposited in or near the **female reproductive tract**, and fertilization occurs within the tract.
 - Internal fertilization requires behavioral interactions and compatible copulatory organs.
 - **All fertilization requires critical timing**, often *mediated by environmental cues, pheromones, and/or courtship behavior.*

Ensuring the Survival of Offspring

- All species produce more offspring than the environment can handle, and the proportion that survives is quite small.
- Species with external fertilization produce more gametes than species with internal fertilization.

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- Species with *internal fertilization* provide *greater protection* of the *embryos* and *more parental care*.
 - The embryos of some *terrestrial animals* develop in *amniote eggs* with protective layers.
 - Some other animals retain the embryo, which develops inside the female.
 - In many animals, parental care helps ensure survival of offspring.

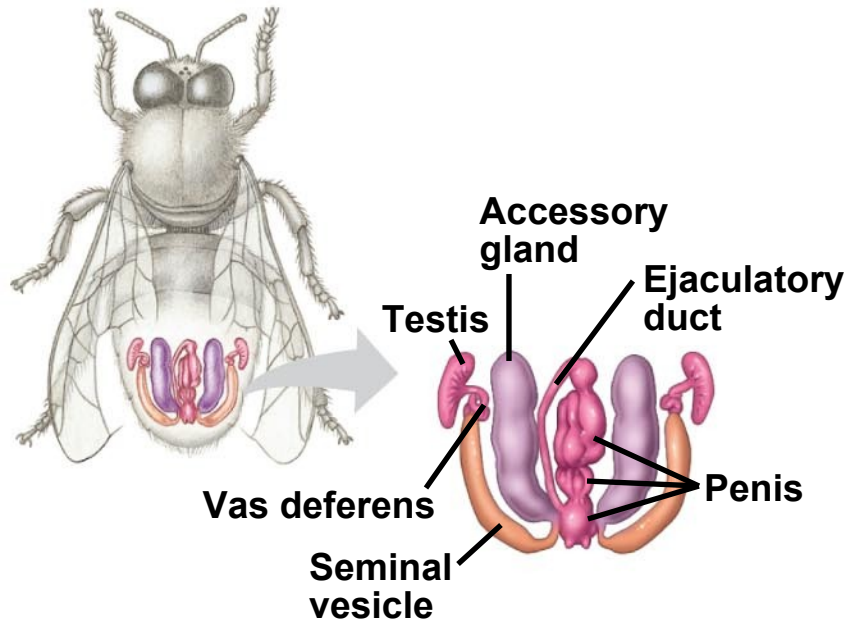
Parental care in an invertebrate



Animal Gamete Production and Delivery

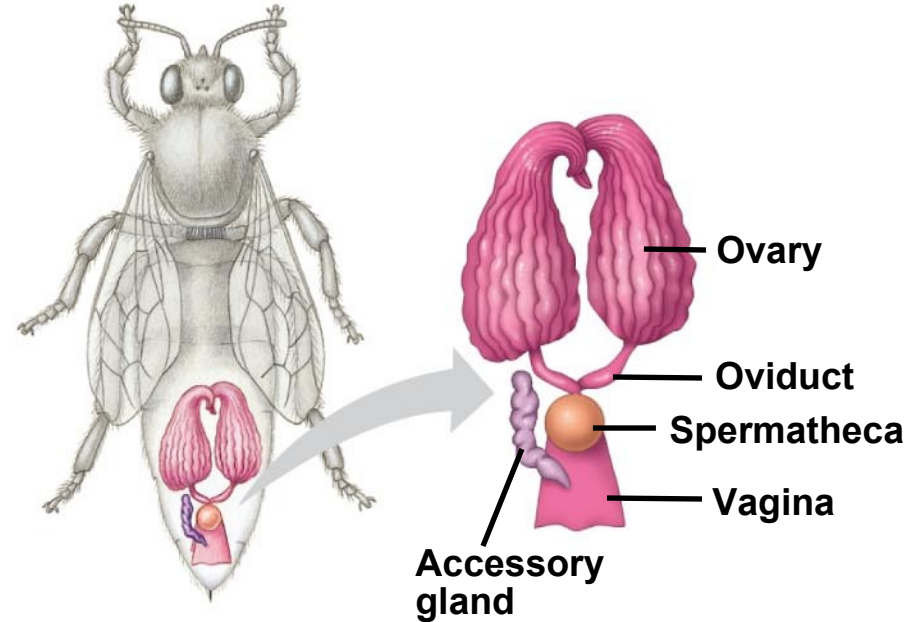
- To reproduce **sexually**, animals must have systems that produce **gametes**.
- In most species individuals have **gonads** = **sex organs** that produce gametes.
- Some simple systems do not have gonads, but gametes form from undifferentiated tissue.
- The most complex systems contain many sets of accessory tubes and glands that carry, nourish, and protect gametes and developing embryos.

Most insects have separate sexes with complex reproductive systems. In many insects, the female has a **spermatheca** in which sperm is stored during copulation.



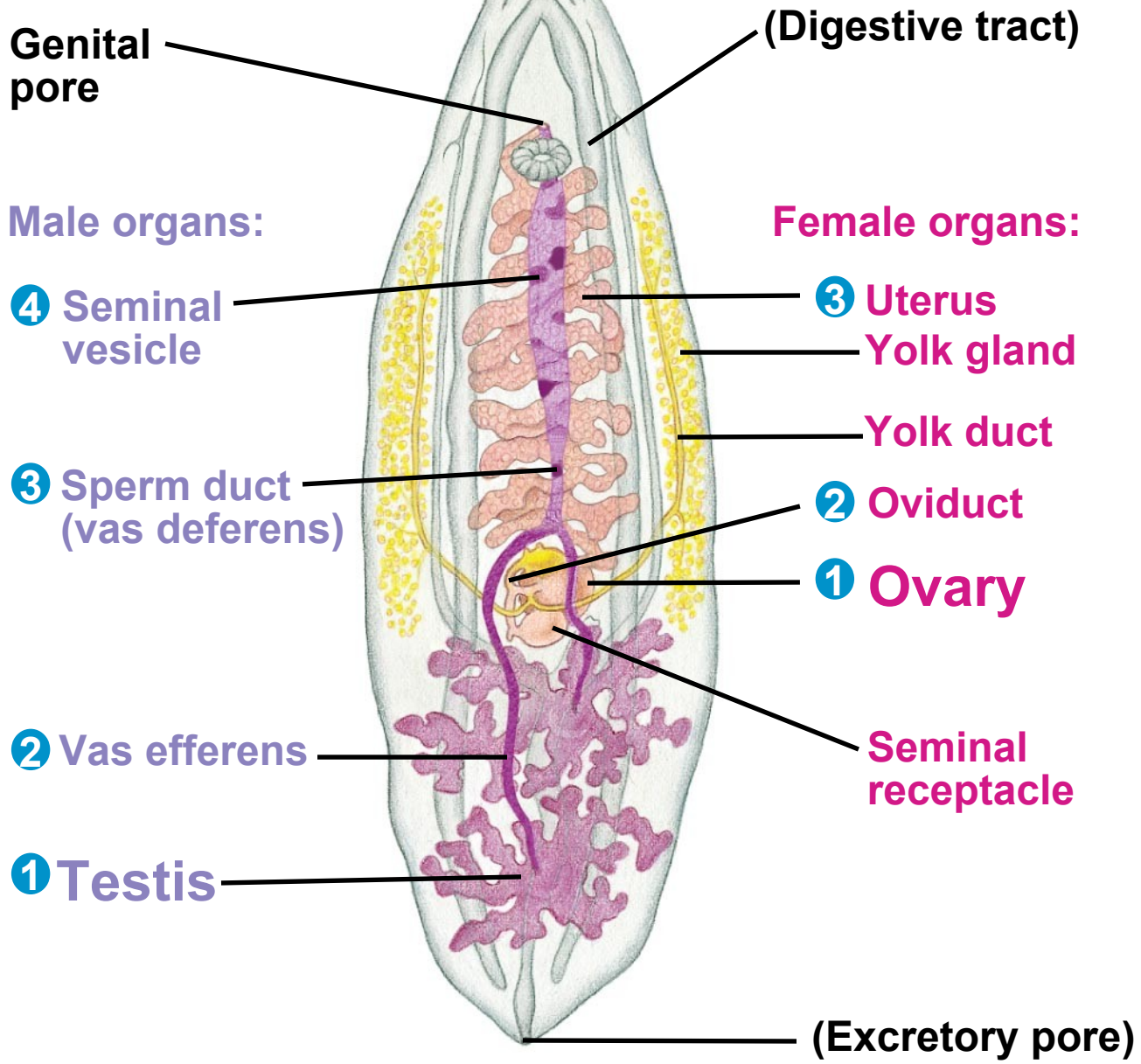
(a) Male honeybee (drone)

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(b) Female honeybee (queen)

Reproductive anatomy
of a **hermaphrodite**



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- A **cloaca** is a common opening between the external environment and the digestive, excretory, and reproductive systems.
 - A cloaca is common in nonmammalian vertebrates; mammals usually have a separate opening to the digestive tract.

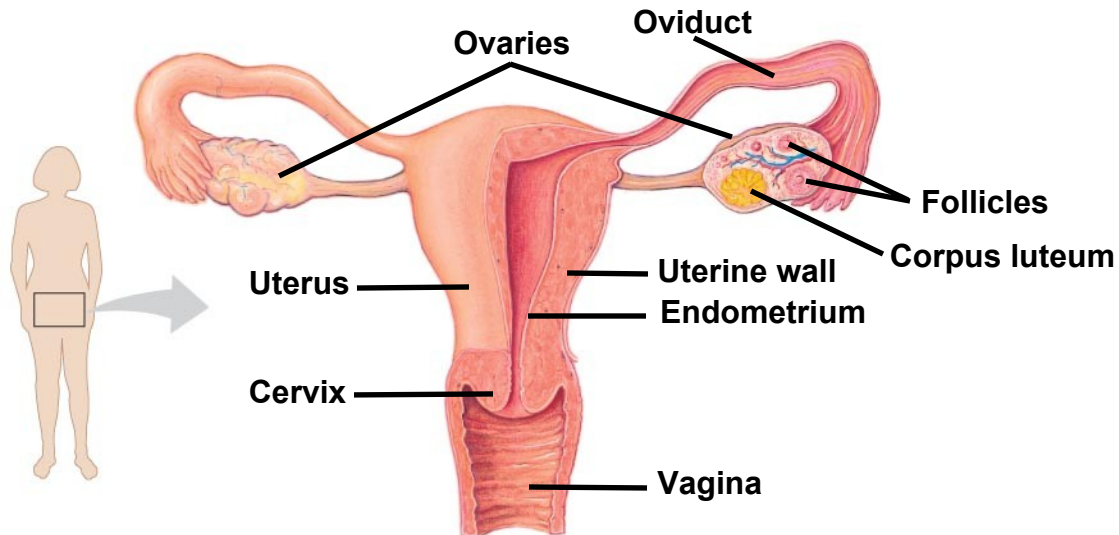
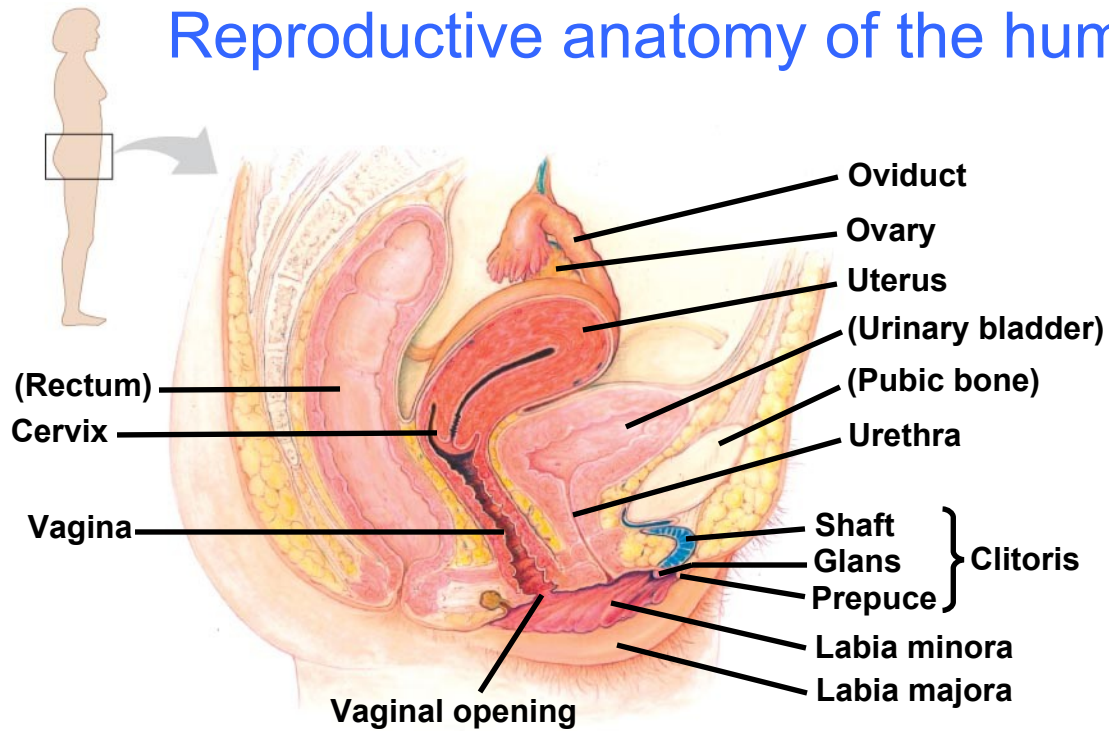
Reproductive organs produce and transport gametes

- The following section focuses on the human reproductive system.

Female Reproductive Anatomy

- The female external reproductive structures include the clitoris and two sets of labia.
- The internal organs are a pair of gonads and a system of ducts and chambers that carry gametes and house the embryo and fetus.

Reproductive anatomy of the human female



Ovaries = Female Gonads

- The female gonads, the ovaries, lie in the abdominal cavity.
- Each ovary contains many **follicles**, which are **egg chambers** consisting of a partially developed egg, called an **oocyte**, surrounded by support cells.
- Once a month, an oocyte develops into an ovum (egg) by the process of **oogenesis**.

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- **Ovulation** expels an egg cell from the follicle.
 - The remaining follicular tissue grows within the ovary, forming a mass called the **corpus luteum**.
 - The corpus luteum secretes hormones that help to maintain pregnancy.
 - If the egg is not fertilized, the corpus luteum degenerates.

Oviducts and Uterus

- The egg cell travels from the ovary to the uterus via an **oviduct**, or **fallopian tube**.
- Cilia in the oviduct convey the egg to the **uterus**, also called the womb.
- The uterus lining, the **endometrium**, has many blood vessels.
- The uterus narrows at the **cervix**, then opens into the **vagina**.

Vagina and Vulva

- The **vagina** is a thin-walled chamber that is the repository for sperm during copulation and serves as the **birth canal**.
- The vagina opens to the outside at the **vulva**, which consists of the **labia majora**, **labia minora**, **hymen**, and **clitoris**.

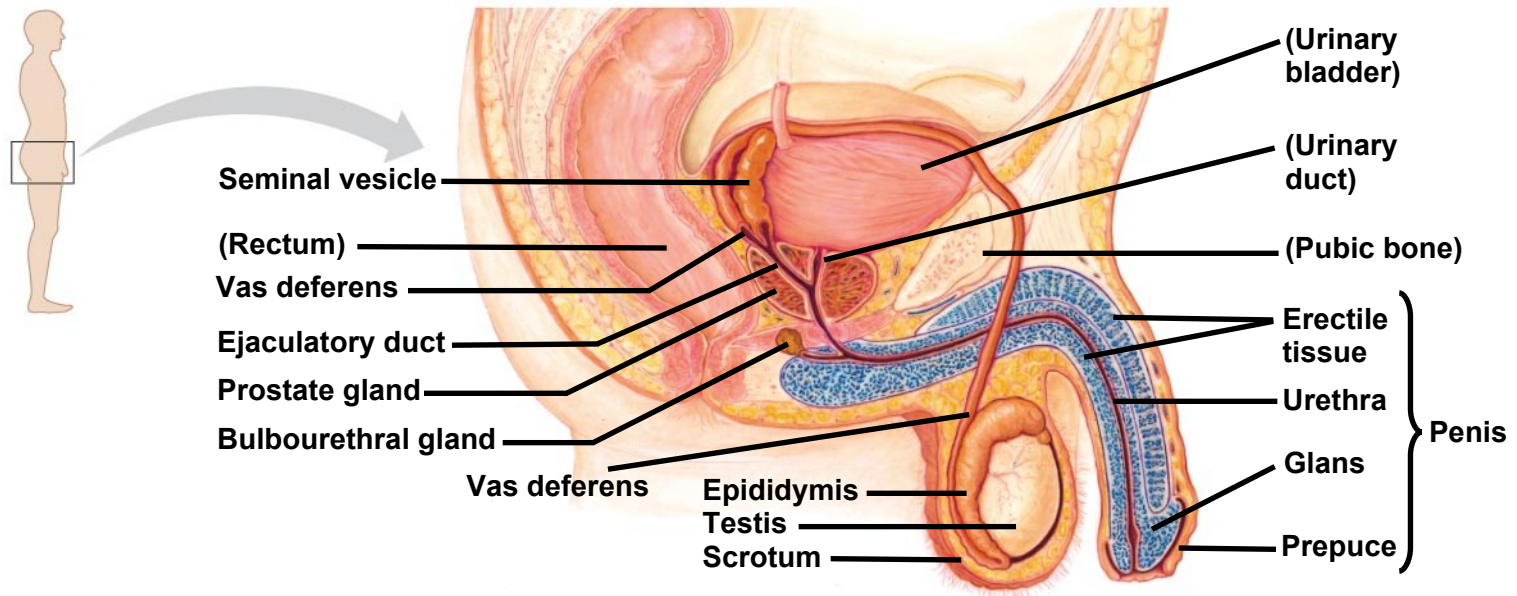
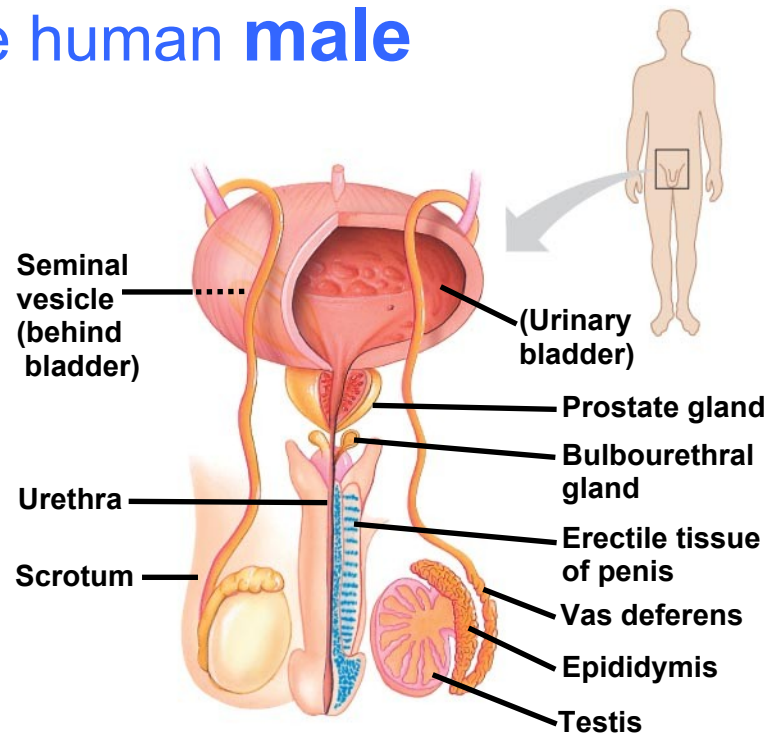
Mammary Glands

- The **mammary glands** are not part of the reproductive system but are important to mammalian reproduction.
- Within the glands, small sacs of epithelial tissue **secrete milk**.

Male Reproductive Anatomy

- The **male's external reproductive organs** are the scrotum and penis.
- Internal organs are the gonads, which produce sperm and hormones, and accessory glands.

Reproductive anatomy of the human male



Testes = Male Gonads

- The **testes** consist of highly coiled tubes surrounded by connective tissue. Sperm form in these **seminiferous tubules**. **Leydig cells** produce hormones and are scattered between the tubules.
- Production of normal sperm cannot occur at the body temperatures of most mammals. So the **testes** are held **outside** the **abdominal cavity** in the **scrotum**, where the **temperature is lower** than in the abdominal cavity.

Ducts

- From the seminiferous tubules of a testis, mature sperm pass into the coiled tubules of the **epididymis**.
- During **ejaculation**, sperm are propelled through the muscular **vas deferens** and the **ejaculatory duct**, and then exit the penis through the **urethra**.

Accessory Glands

- **Semen** is composed of sperm plus secretions from three sets of accessory glands.
- The two **seminal vesicles** contribute about 60% of the total volume of semen.
- The **prostate gland** secretes its products directly into the urethra through several small ducts.
- The *bulbourethral glands* secrete a clear mucus before ejaculation that neutralizes acidic urine remaining in the urethra.

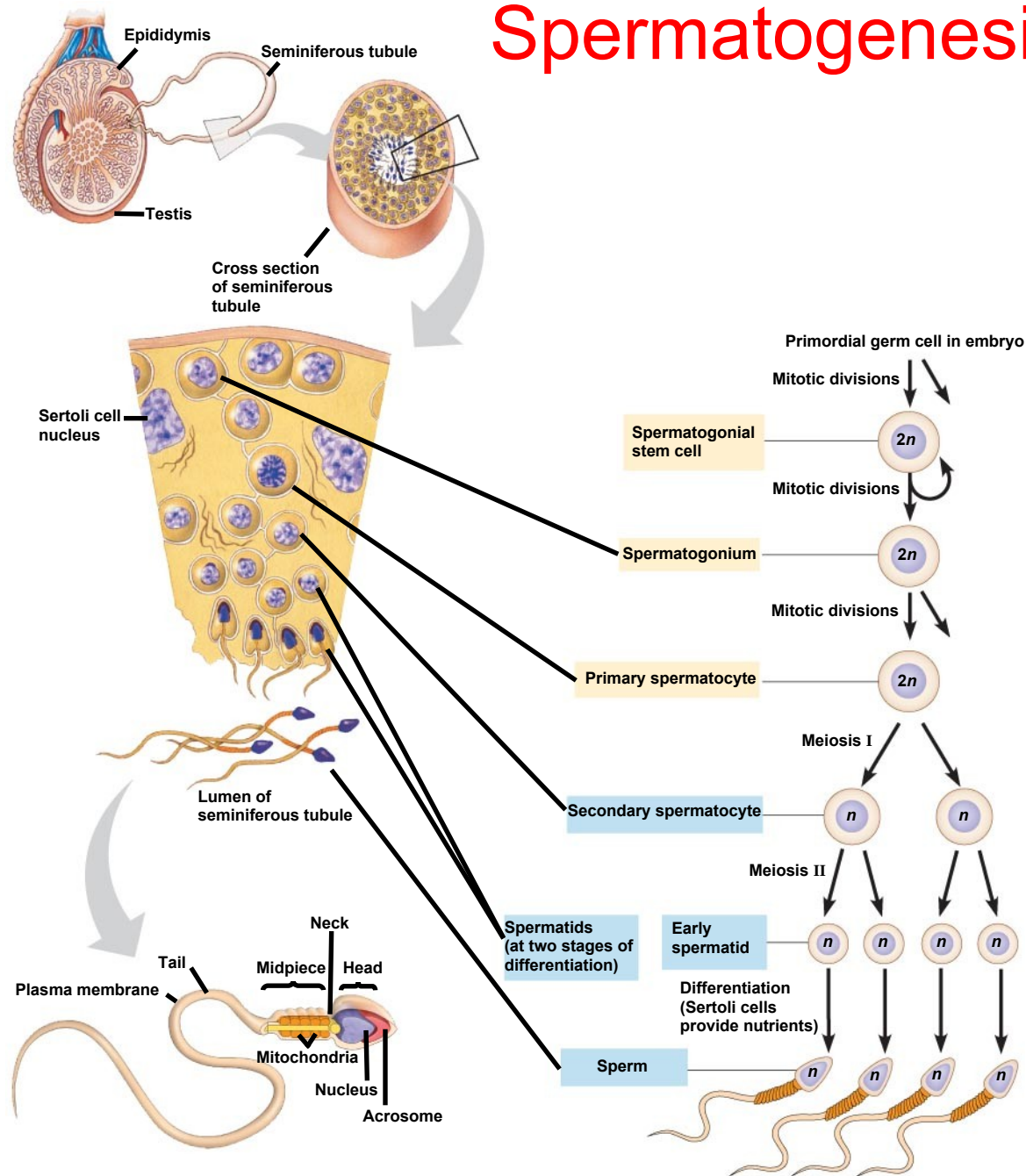
Penis

- The human ***penis*** is composed of three cylinders of *spongy erectile tissue*.
- During sexual arousal, the erectile tissue fills with blood from the arteries, causing an erection.
- The head of the penis has a thinner skin covering than the shaft, and is more sensitive to stimulation.

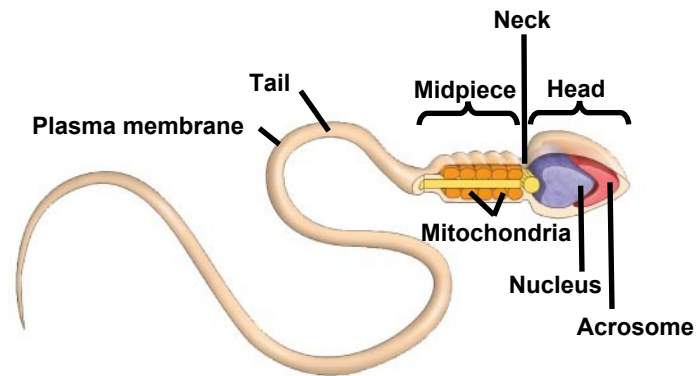
The timing and pattern of **meiosis** in mammals differ for males and females

- **Gametogenesis** = the production of gametes by meiosis. This differs in females and males
- Sperm are small and motile and are produced throughout the life of a sexually mature male.
- **Spermatogenesis** is production of mature sperm.

Spermatogenesis



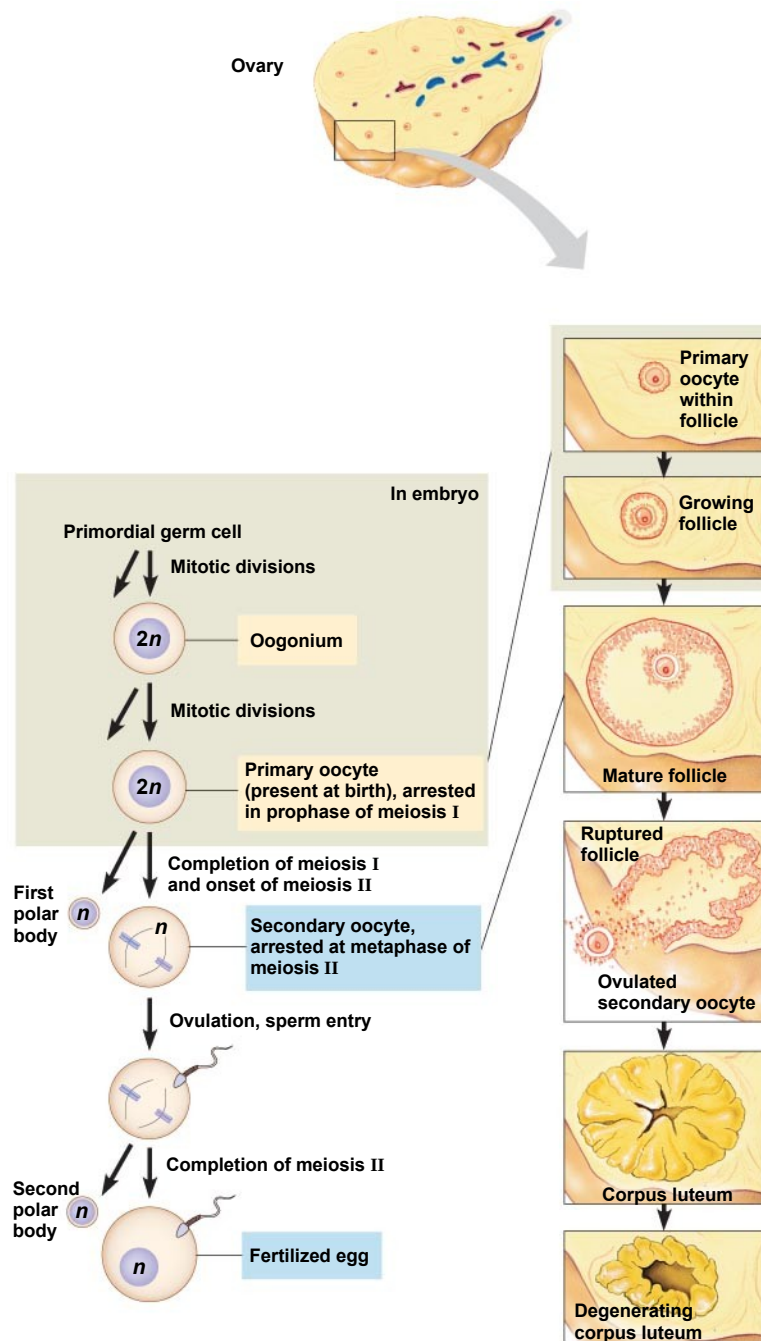
Mature sperm



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- Eggs contain stored nutrients and are much larger.
 - **Oogenesis** is development of **mature oocytes** (**eggs**) and can take many years .

Oogenesis



Spermatogenesis vs. Oogenesis

- Spermatogenesis differs from oogenesis:
 - In oogenesis, one egg forms from each cycle of meiosis; in spermatogenesis four sperm form from each cycle of meiosis.
 - Oogenesis ceases later in life in females; spermatogenesis continues throughout the adult life of males.
 - Oogenesis has long interruptions; spermatogenesis produces sperm from precursor cells in a continuous sequence.

The interplay of **tropic** and **sex hormones** regulates mammalian reproduction

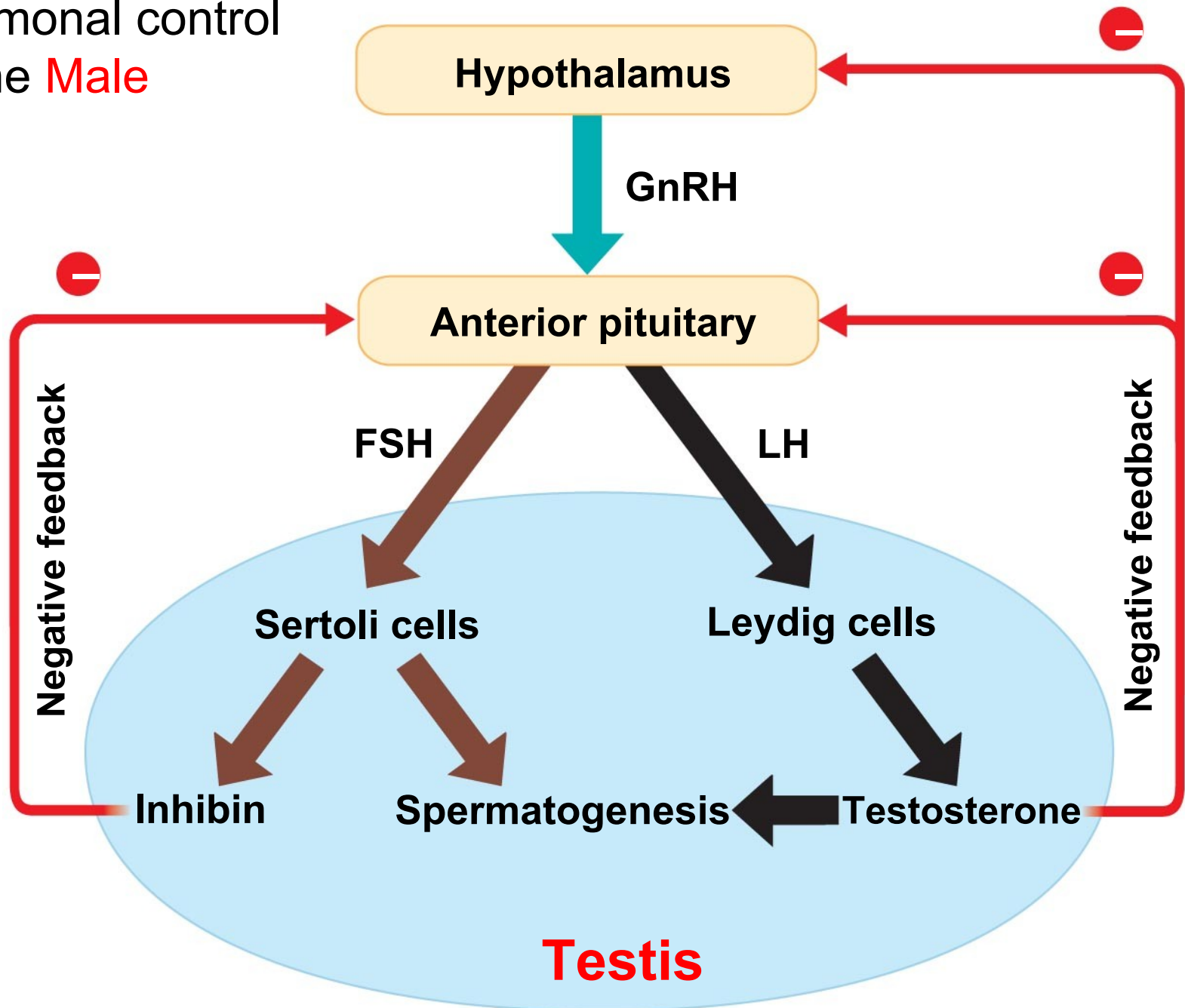
- *Human reproduction is coordinated by **hormones** from the **hypothalamus**, **anterior pituitary**, and **gonads**.*
- Gonadotropin-releasing hormone (**GnRH**) is secreted by the **hypothalamus** and directs the release of FSH and LH from the **anterior pituitary**.
- **FSH** and **LH** regulate processes in the **gonads** and the production of **sex hormones**.

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- The **sex hormones** are androgens, estrogens, and progesterone.
 - **Sex hormones regulate:**
 - The development of **primary sex characteristics** during embryogenesis
 - The development of **secondary sex characteristics** at puberty
 - **Sexual behavior** and sex drive.

Hormonal Control of the **Male** Reproductive System

- **FSH** promotes the activity of **Sertoli cells**, which nourish developing sperm and are located within the **seminiferous tubules**.
- **LH** regulates **Leydig cells**, which secrete testosterone and other androgen hormones, which in turn promote spermatogenesis.

Hormonal control in the **Male**



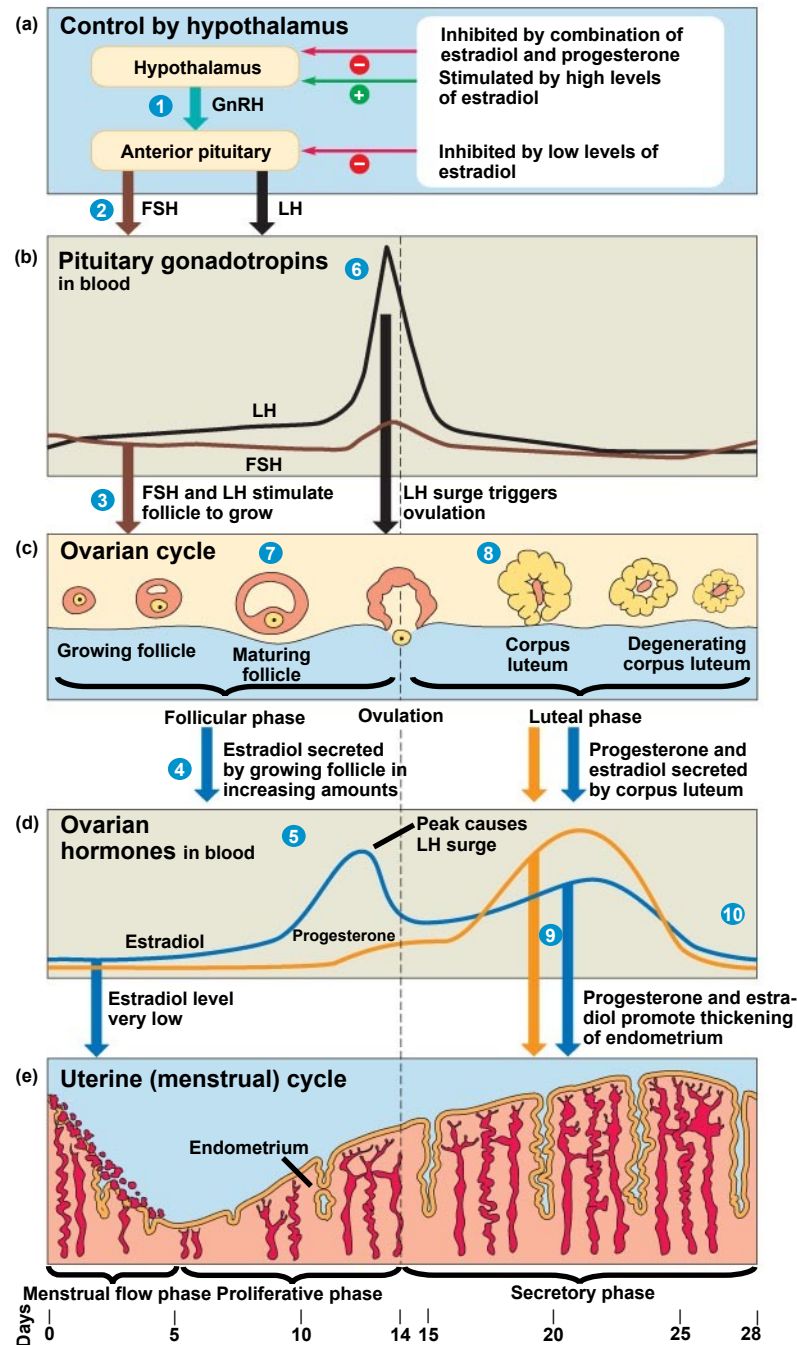
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- Testosterone regulates the production of GnRH, FSH, and LH through **negative feedback mechanisms**.
 - Sertoli cells secrete the hormone **inhibin**, which reduces FSH secretion from the anterior pituitary.

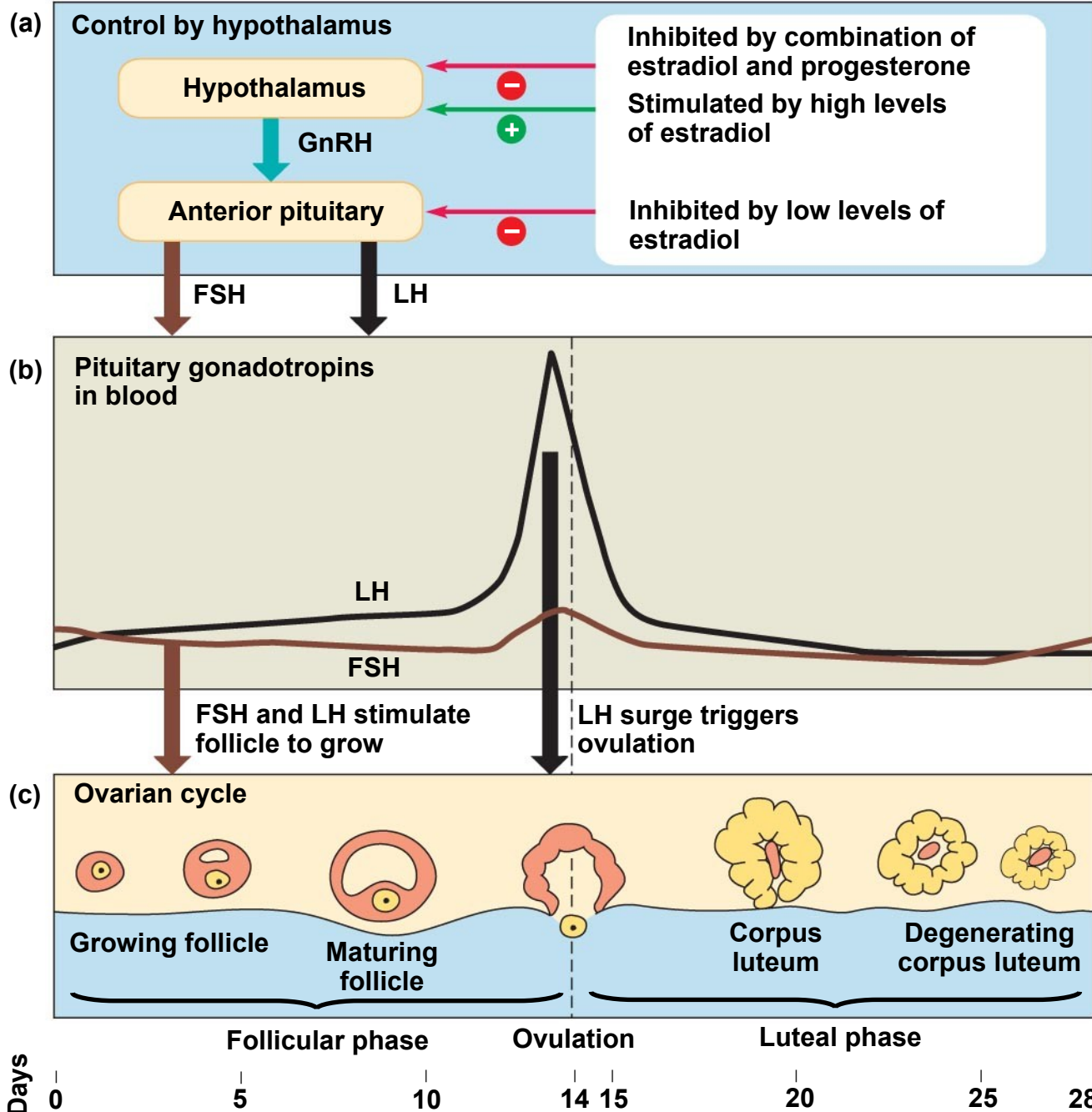
The Reproductive Cycles of Females

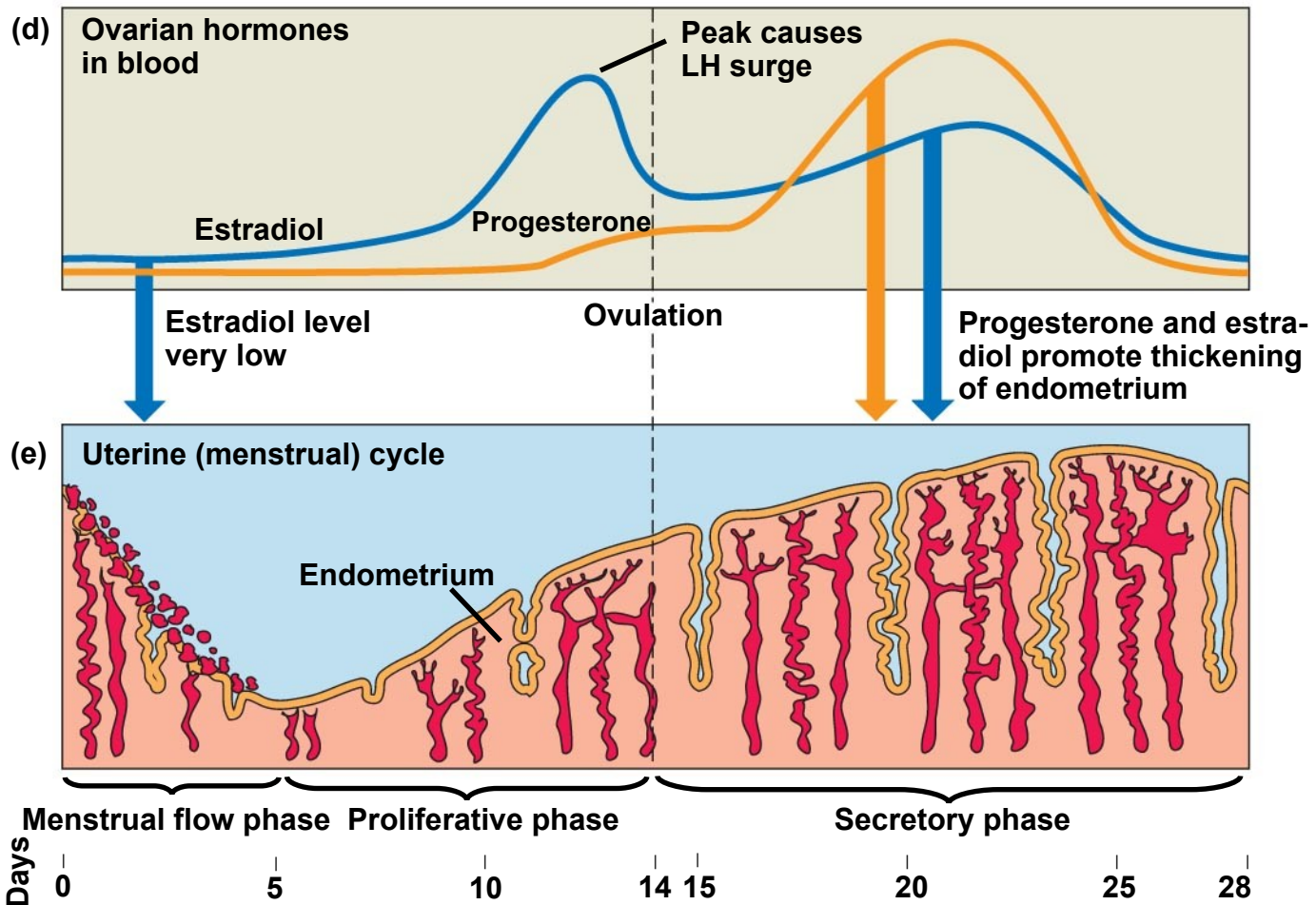
- In females, the secretion of hormones and the **reproductive events** they regulate are **cyclic**.
- Prior to **ovulation**, the **endometrium** = uterine lining, **thickens** with blood vessels in preparation for embryo implantation.
- If an embryo does not implant in the endometrium, the endometrium is shed in a process called **menstruation**.

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- Hormones closely link the **two cycles** of female reproduction:
 - Changes in the uterus / uterine lining with blood vessels define the **menstrual cycle** (also called the **uterine cycle**).
 - Changes in the ovaries / follicle / egg chamber define the **ovarian cycle**.

The reproductive cycle of the human female







The Ovarian Cycle

- The sequential release of GnRH then FSH and LH stimulates **follicle growth**.
- Follicle growth and an increase in the hormone estradiol characterize the **follicular phase** of the ovarian cycle.
- The follicular phase **ends at ovulation**, and the secondary **oocyte is released**.

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- Following ovulation, the **follicular tissue** left behind **transforms into** the **corpus luteum**; this is the **luteal phase**.
 - The corpus luteum disintegrates, and ovarian steroid hormones decrease .

The Uterine (Menstrual) Cycle

- **Hormones coordinate the *uterine cycle* with the *ovarian cycle*:**
 - **Thickening** of the **endometrium** during the **proliferative phase** coordinates with the follicular phase.
 - Secretion of nutrients during the **secretory phase** coordinates with the luteal phase.
 - **Shedding of the endometrium** during the **menstrual flow phase** coordinates with the growth of new ovarian follicles.

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- A new cycle begins if no embryo implants in the endometrium.
 - Cells of the uterine lining can sometimes migrate to an abnormal, or **ectopic**, location.
 - Swelling of these cells in response to hormone stimulation results in a disorder called **endometriosis**.

Menopause

- After about 500 cycles, human females undergo **menopause**, the **cessation of ovulation and menstruation**.
- Menopause is very unusual among animals.
- Menopause might have evolved to allow a mother to provide better care for her children and grandchildren.

Menstrual vs Estrous Cycles

- Menstrual cycles are characteristic of humans and some other primates:
 - The endometrium is shed from the uterus in a bleeding called menstruation
 - Sexual receptivity is not limited to a timeframe.

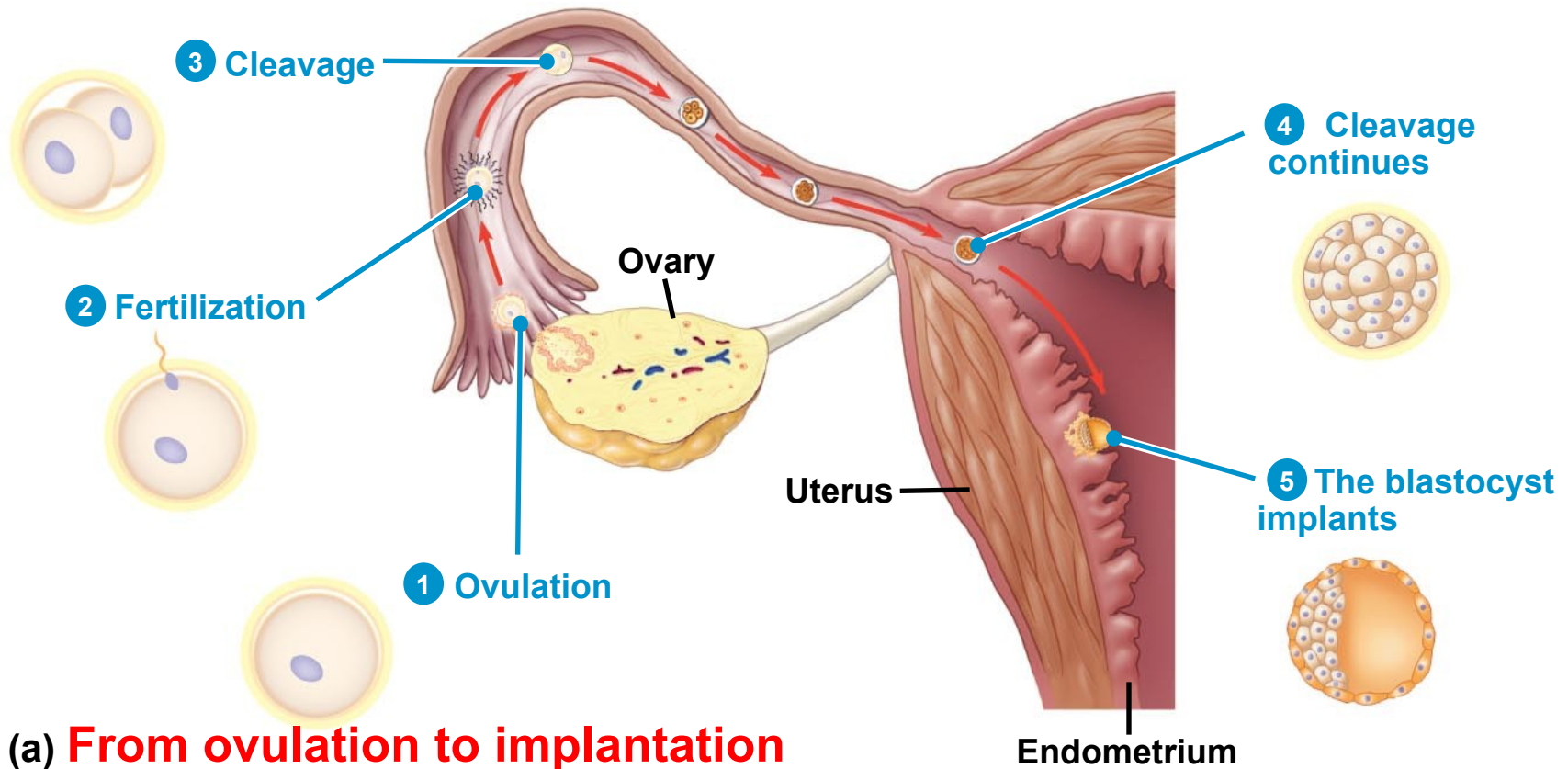
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- **Estrous cycles** are characteristic of most mammals:
 - The endometrium is reabsorbed by the uterus
 - Sexual receptivity is limited to a “heat” period
 - The length and frequency of estrus cycles varies from species to species.

In placental mammals, an embryo develops fully within the mother's uterus

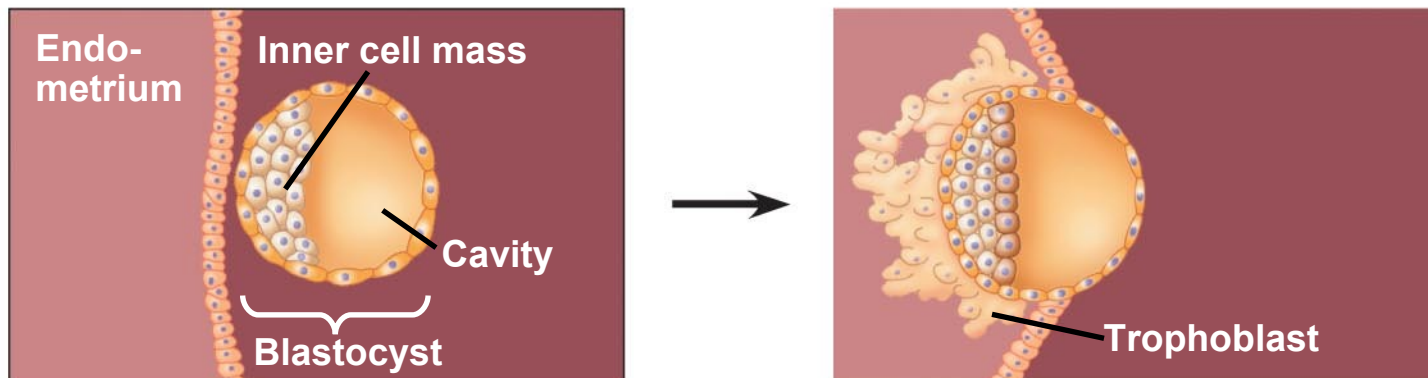
- An egg develops into an embryo in a series of predictable events.

Conception, Embryonic Development, and Birth

- **Conception** = fertilization of an egg by a sperm, occurs in the oviduct.
- The resulting zygote begins to divide by mitosis in a process called cleavage.
- Division of cells gives rise to a blastocyst, a ball of cells with a cavity.



(a) From ovulation to implantation



(b) Implantation of blastocyst

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- After blastocyst formation, the embryo implants into the endometrium.
 - The embryo releases **human chorionic gonadotropin (hCG)**, which prevents menstruation.
 - **Pregnancy**, or **gestation**, is the condition of carrying one or more embryos in the **uterus**.
 - Duration of pregnancy in other species correlates with body size and maturity of the young at birth.

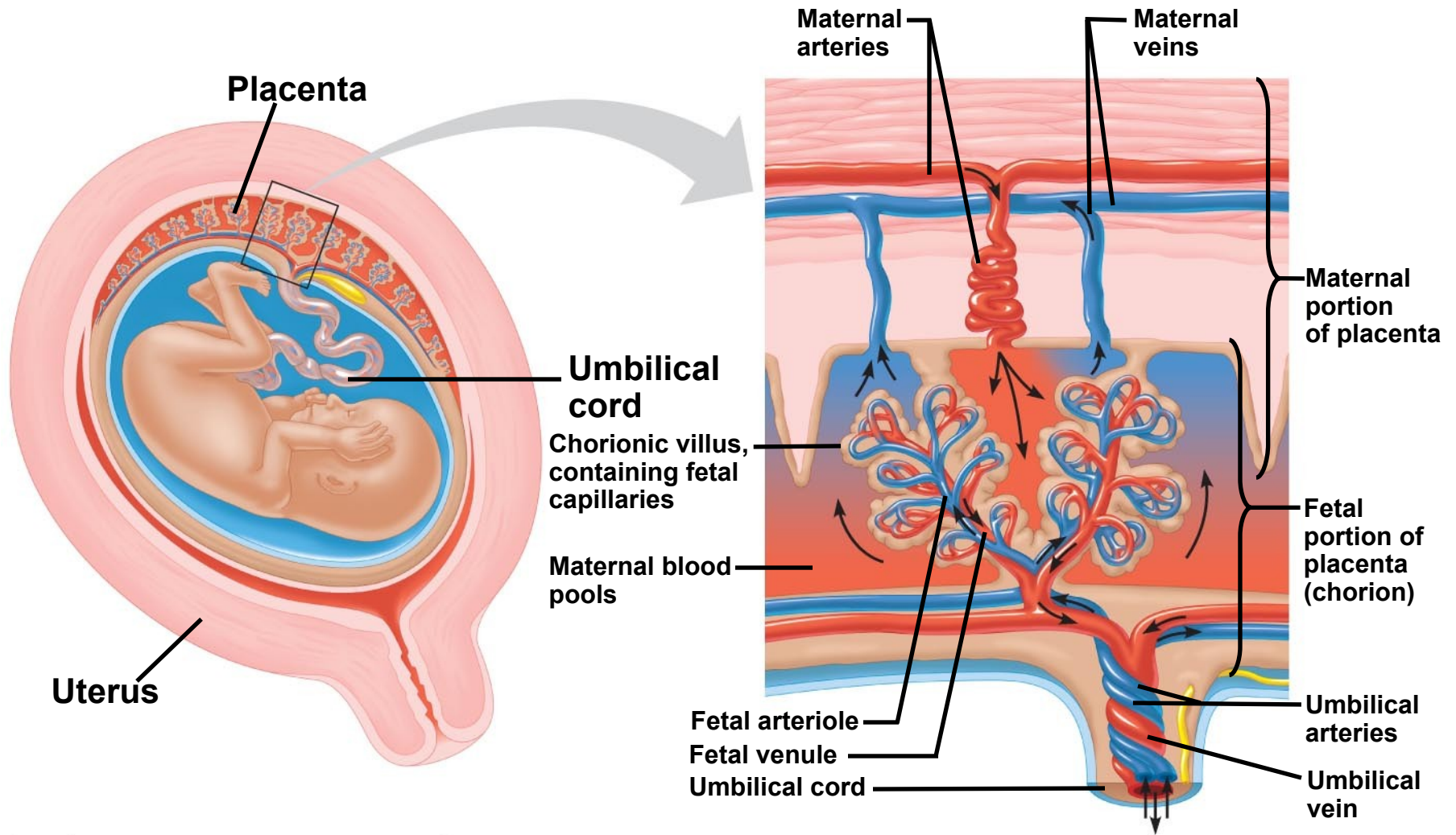
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- Pregnancies can terminate spontaneously due to chromosomal or developmental abnormalities.
 - An ectopic pregnancy occurs when a fertilized egg begins to develop in the fallopian tube.

First Trimester

- Human gestation can be divided into three **trimesters** of about three months each.
- The first trimester is the time of most radical change for both the mother and the embryo.
- During implantation, the endometrium grows over the blastocyst.

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- During its first 2 to 4 weeks, the embryo obtains nutrients directly from the endometrium.
 - Meanwhile, the **outer layer** of the **blastocyst**, called the **trophoblast**, mingles **with** the **endometrium** and eventually **forms** the **placenta**.
 - Blood from the embryo travels to the placenta through arteries of the umbilical cord and returns via the umbilical vein.

Placental circulation



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- Splitting of the embryo during the first month of development results in genetically **identical twins**. Release and fertilization of two eggs results in **fraternal** and genetically distinct twins.
 - The first trimester is the main period of **organogenesis** = development of the body organs.
 - All the major structures are present by 8 weeks, and the embryo is called a **fetus**.

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- Changes occur in the mother:
 - Growth of the placenta
 - Cessation of ovulation and the menstrual cycle
 - Breast enlargement
 - Nausea is also very common.

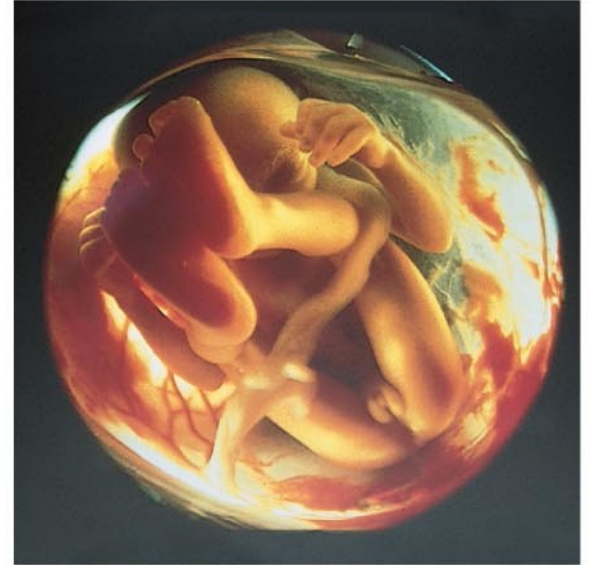
Human fetal development



(a) 5 weeks

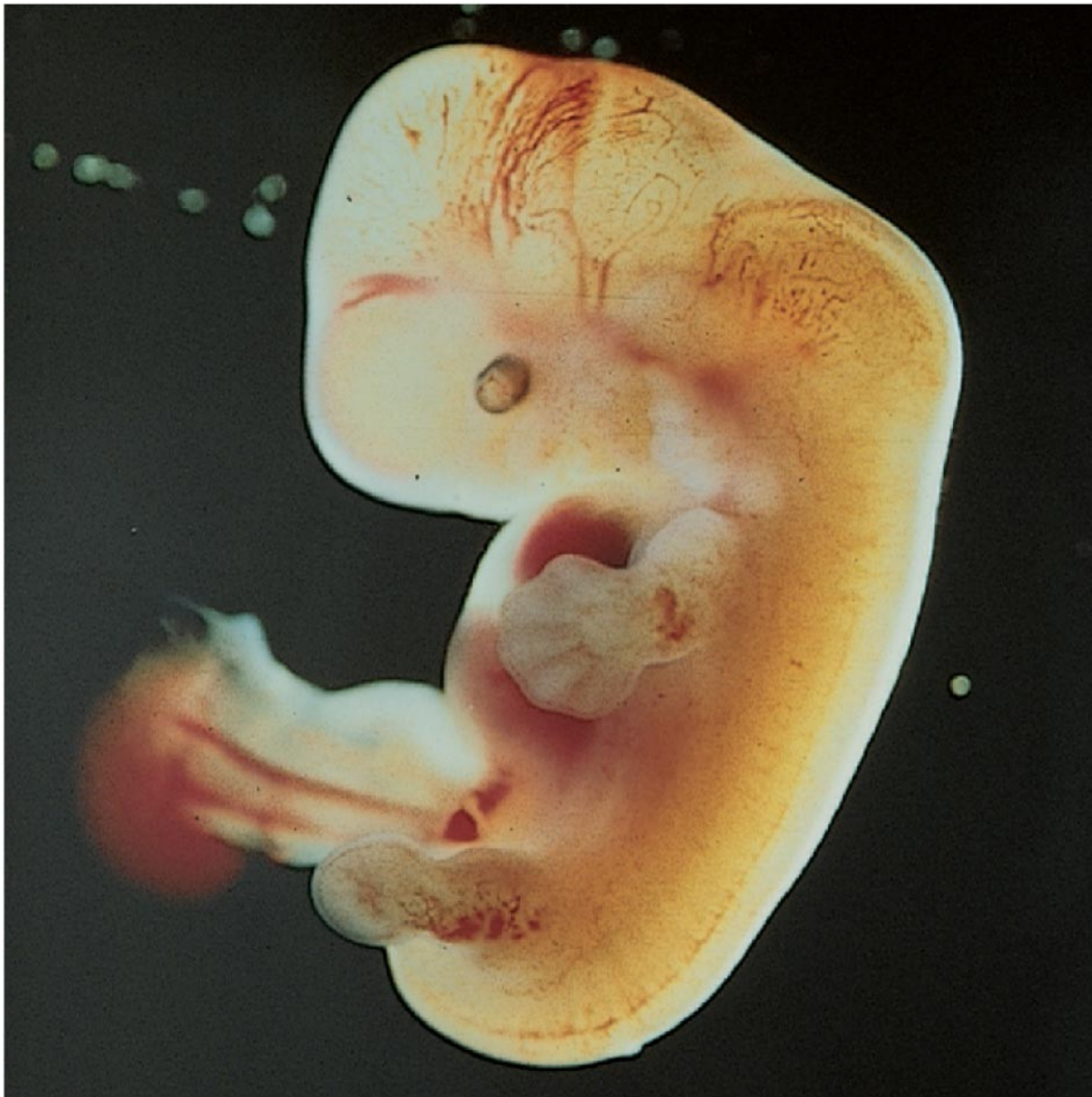


(b) 14 weeks



(c) 20 weeks

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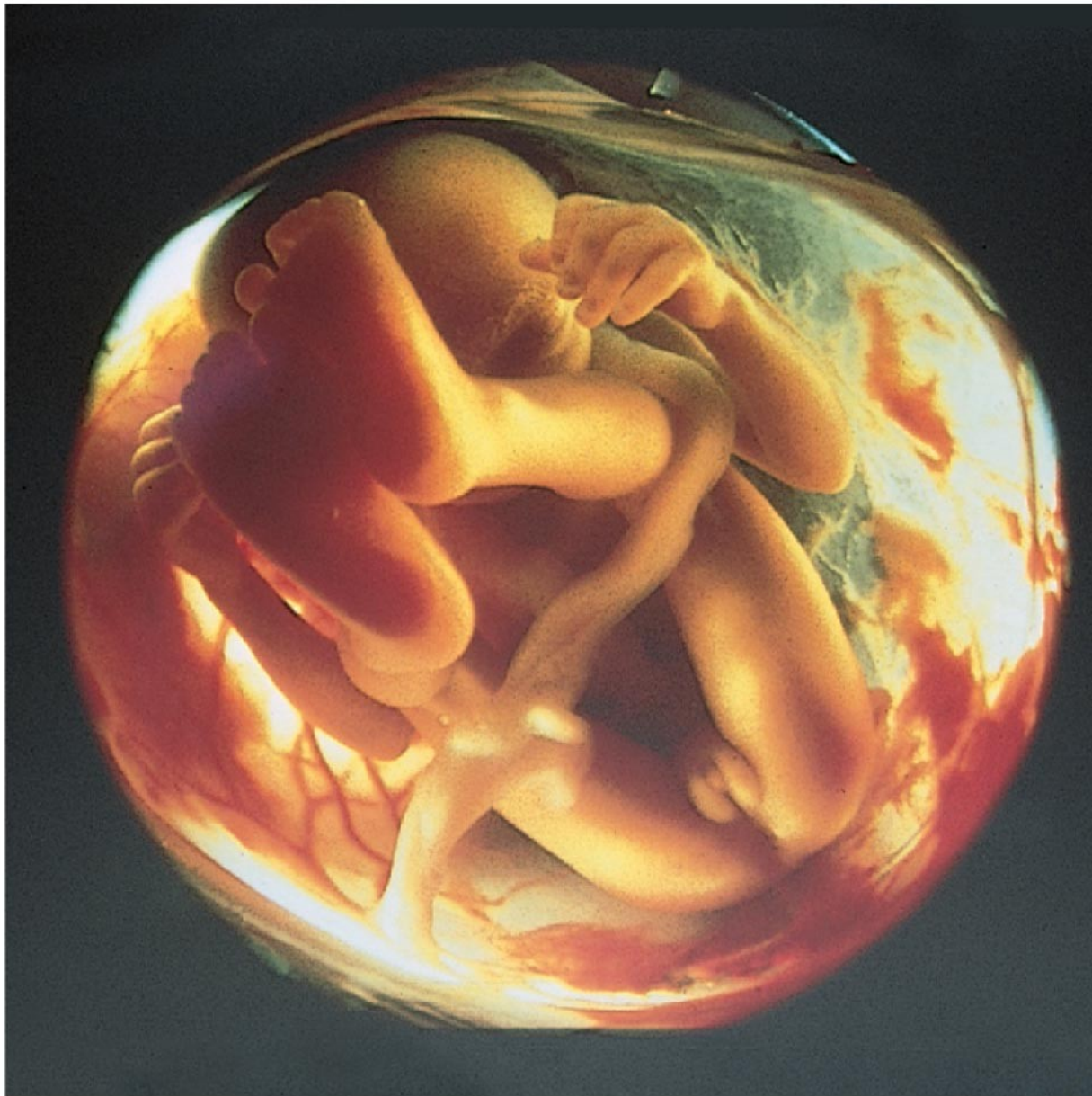


(a) 5 weeks



(b) 14 weeks

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(c) 20 weeks

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Second Trimester

- During the second trimester:
 - The fetus grows and is very active
 - The mother may feel fetal movements
 - The uterus grows enough for the pregnancy to become obvious.

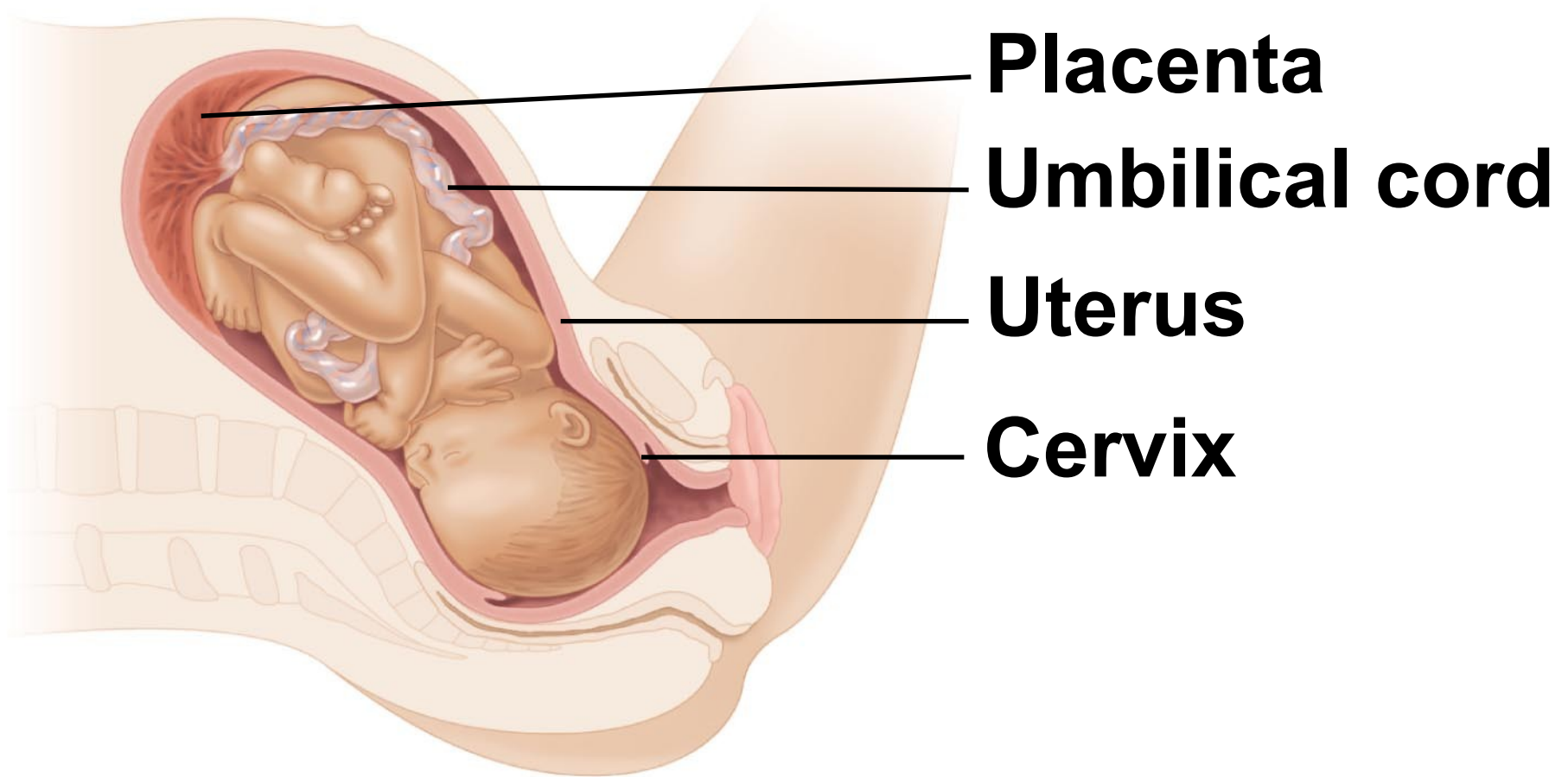
Third Trimester

- During the third trimester, the fetus grows and fills the space within the embryonic membranes.
- A complex interplay of local regulators and hormones induces and regulates **labor**, the process by which childbirth occurs.

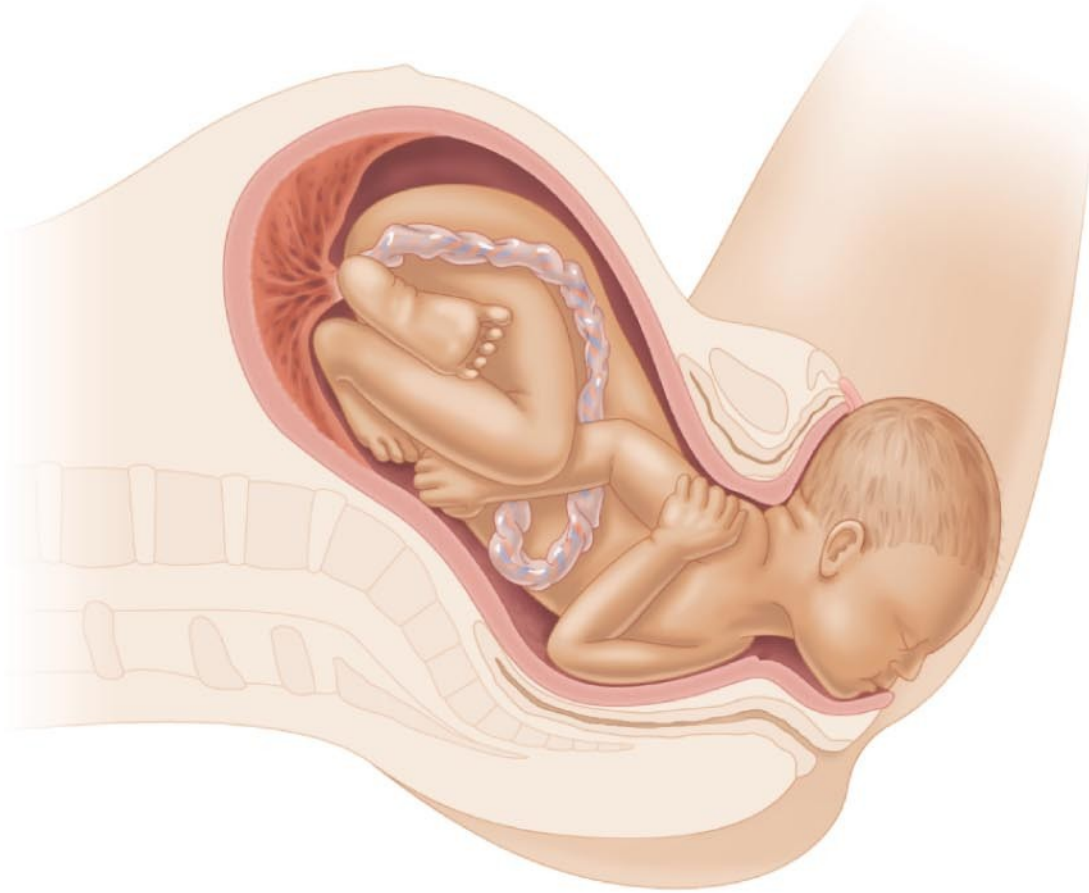
The diagram illustrates the positive feedback loop for childbirth. It begins with **Estradiol** (from ovaries) inducing **oxytocin receptors on uterus**. **Oxytocin** (from fetus and mother's posterior pituitary) then stimulates the uterus to contract and stimulates the placenta to make **Prostaglandins**. Prostaglandins stimulate more contractions of the uterus, which in turn stimulates the placenta to make more prostaglandins, creating a cycle. The diagram also shows a **Positive feedback** loop where the stimulation of the uterus to contract leads to more contractions, which further stimulates the uterus to contract.

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graph TD; Estradiol[Estradiol] -- "from ovaries" --> Induces[Induces oxytocin receptors on uterus]; Induces --> Uterus[Uterus]; Oxytocin[Oxytocin] -- "from fetus and mother's posterior pituitary" --> StimUterus[Stimulates uterus to contract]; StimUterus --> Uterus; StimUterus --> StimPlacenta[Stimulates placenta to make Prostaglandins]; StimPlacenta --> Prostaglandins[Prostaglandins]; Prostaglandins --> StimContractions[Stimulate more contractions of uterus]; StimContractions --> Uterus; Uterus -- "Positive feedback" --> StimUterus; Uterus -- "Positive feedback" --> StimPlacenta; Uterus -- "Positive feedback" --> StimContractions;
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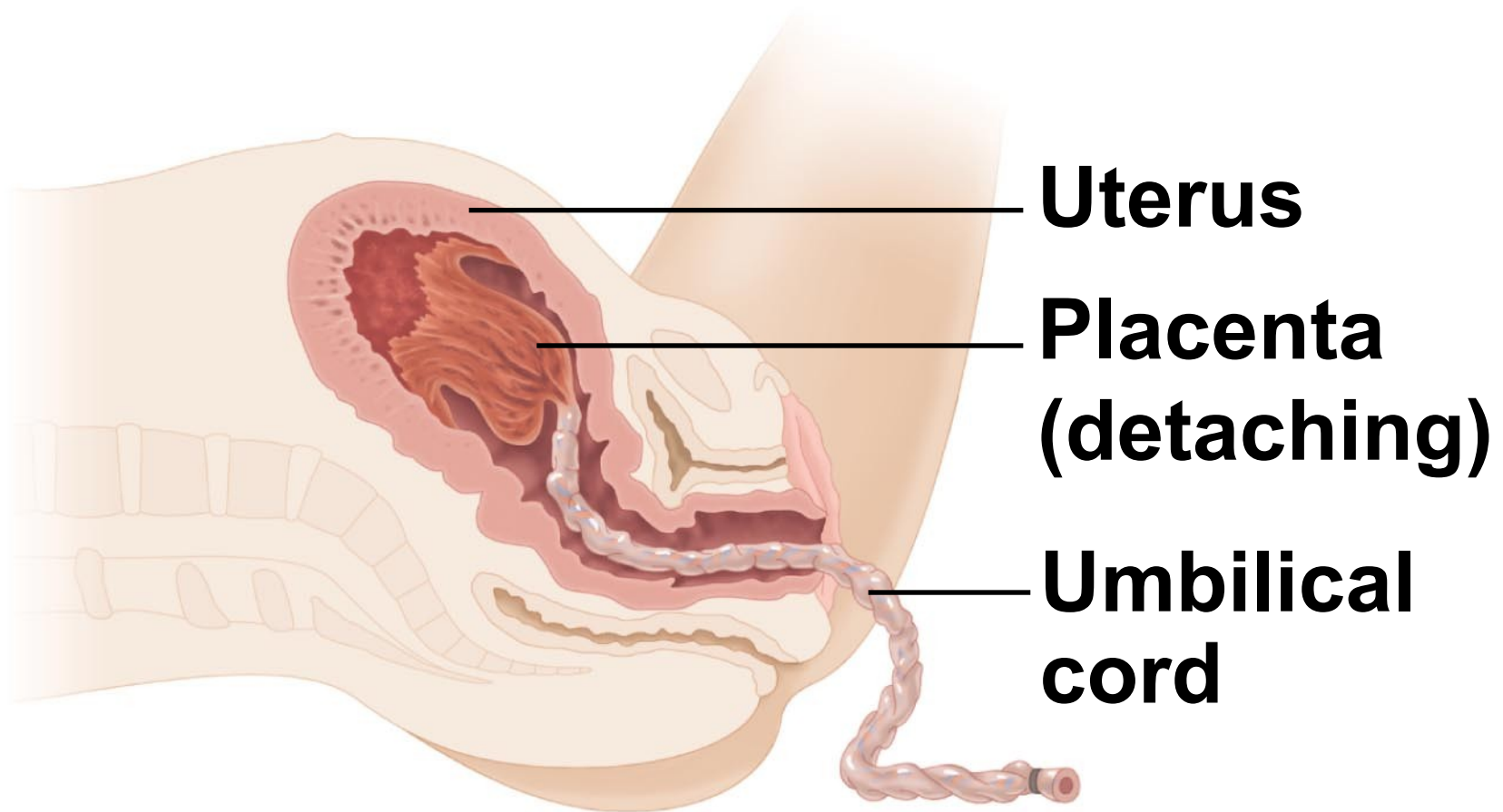
The three stages of labor



1 Dilation of the cervix

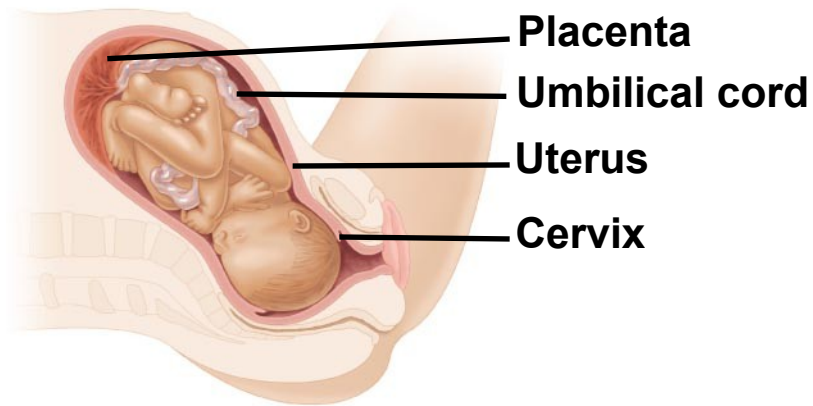


2 Expulsion: delivery of the infant



3 Delivery of the placenta

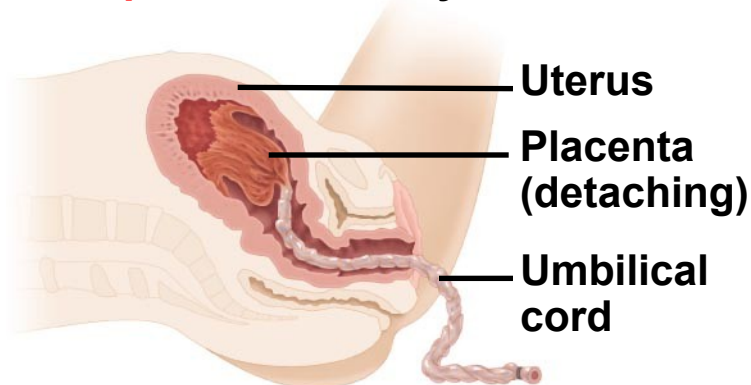
The three stages of labor



1 **Dilation** of the cervix



2 **Expulsion**: delivery of the infant



3 **Delivery** of the placenta

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- **Birth**, or **parturition**, is brought about by a series of strong, **rhythmic uterine contractions**.
 - First the baby is delivered, and then the placenta.
 - **Lactation** = **the production of milk**. This is unique to mammals.

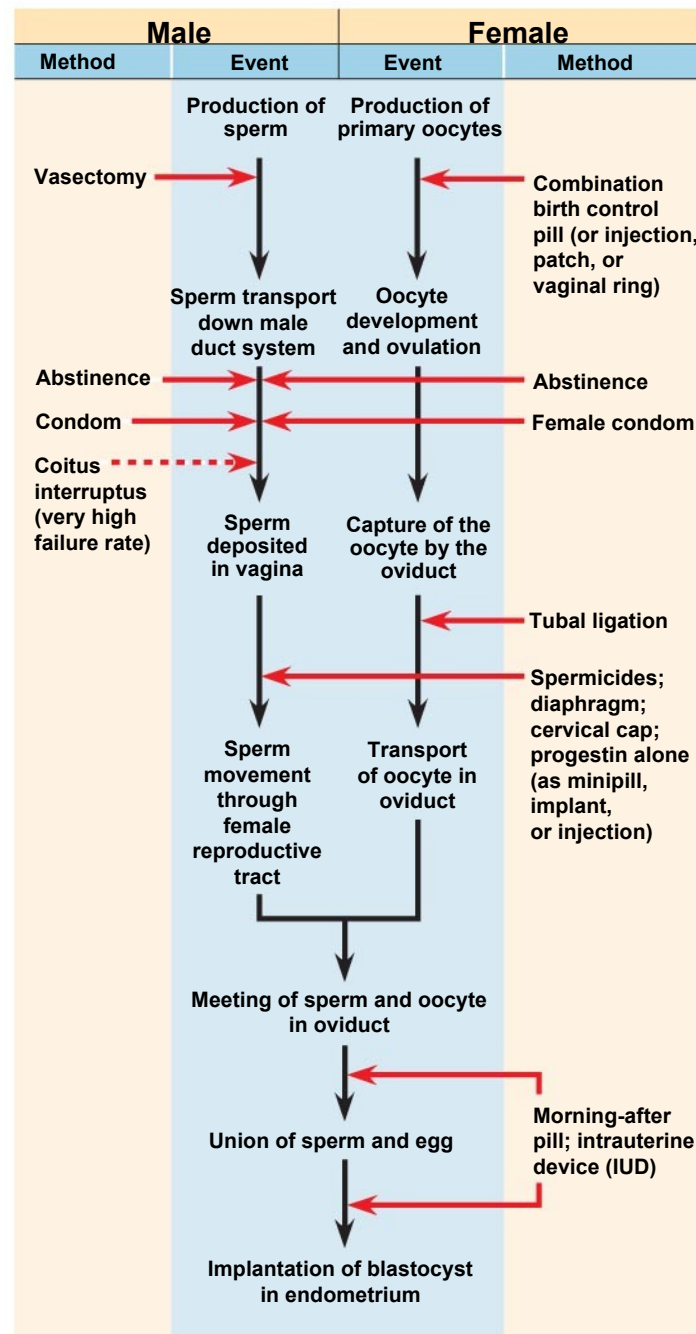
Maternal Immune Tolerance of the Embryo and Fetus

- A woman's acceptance of her "foreign" offspring is not fully understood.
- It may be due to suppression of the immune response in her uterus.

Contraception and Abortion

- **Contraception**, the deliberate prevention of pregnancy, can be achieved in a number of ways.
- Contraceptive methods fall into three categories:
 - Preventing release of eggs and sperm
 - Keeping sperm and egg apart
 - Preventing implantation of an embryo.

Mechanisms of several contraceptive methods



Detecting Disorders During Pregnancy

- **Amniocentesis** and **chorionic villus sampling** are invasive techniques in which amniotic fluid or **fetal cells** are obtained for **genetic analysis**.
- Noninvasive procedures usually use ultrasound imaging to detect fetal condition.
- Genetic testing of the fetus poses ethical questions and can present parents with difficult decisions.

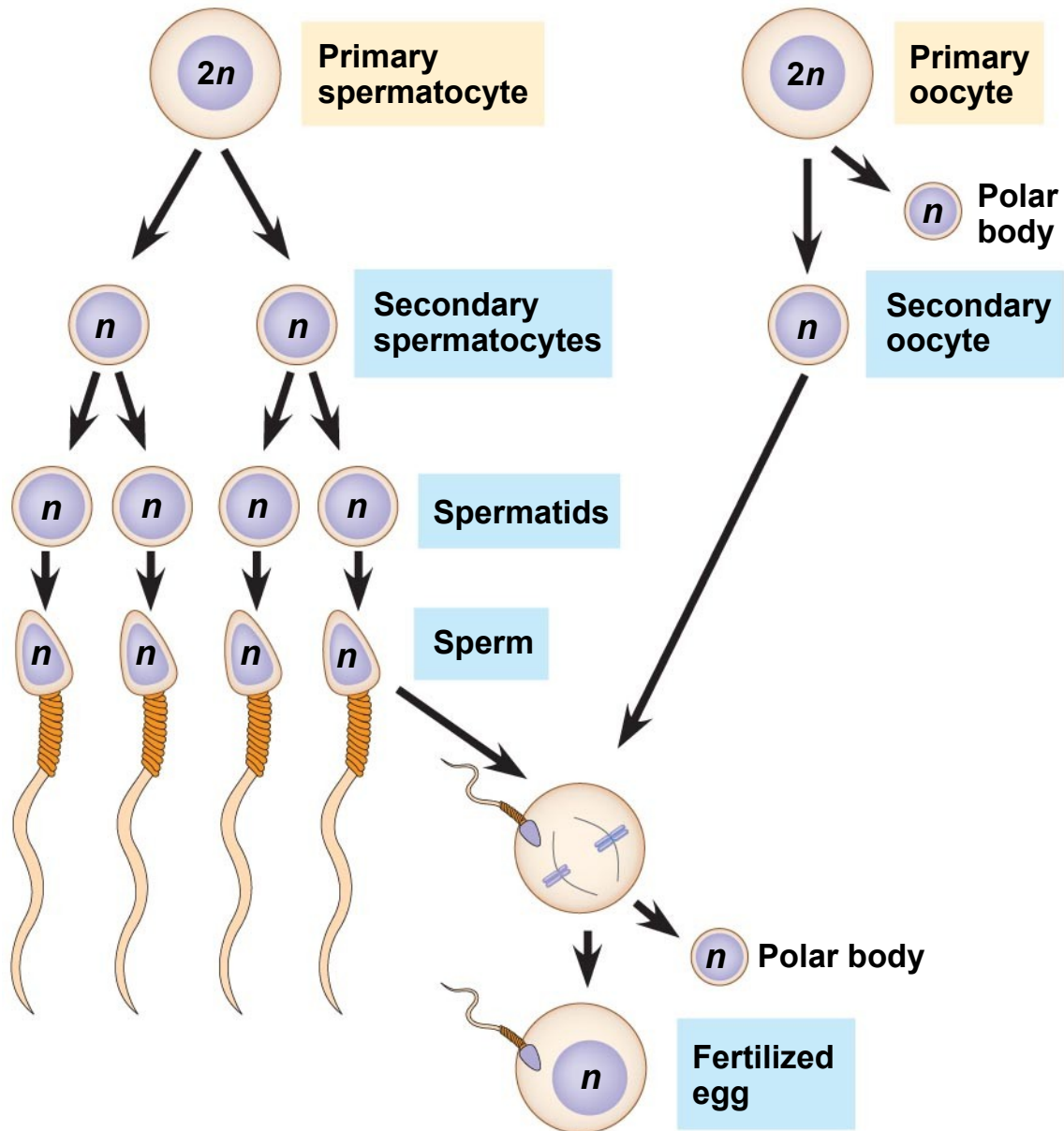
Treating Infertility

- Modern technology can provide infertile couples with **assisted reproductive technologies**.
- ***In vitro fertilization* (IVF)** mixes eggs with sperm in culture dishes and returns the embryo to the uterus at the 8 cell stage.
- Sperm are injected directly into an egg in a type of IVF called **intracytoplasmic sperm injection (ICSI)**.

Gametogenesis

Spermatogenesis

Oogenesis



You should now be able to:

1. Distinguish between asexual and sexual reproduction.
 2. Explain how hermaphroditism may be advantageous to animals that have difficulty encountering a member of the opposite sex.
 3. Describe various ways in which animals may protect developing embryos.
 4. Using diagrams, identify and state the function of each component of the male and female reproductive systems.
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5. Describe oogenesis and spermatogenesis; describe three major differences between them.
 6. Explain how the uterine and ovarian cycles are synchronized and describe the functions of the hormones involved.
 7. List the various methods of contraception, how each works.
 8. Describe techniques that allow us to learn about the health and genetics of a fetus.